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MDM-NER: A multiple dependency modeling driven named entity recognition approach for judicial documents

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ABSTRACT

The characteristic of nested entities spanning a wide range in judicial documents poses significant challenges for entity recognition tasks. This paper proposes multiple dependency modeling driven named entity recognition model (MDM-NER), which can capture the association relationships between characters and words through the encoder module integrating Multi-Head Attention (MHA) and Cross-Attention (CA), and realize multi-dimensional collaborative recognition and label sequence optimization of nested entities through the decoder module composed of a joint predictor and Conditional Random Field model (CRF). It has demonstrated better comprehensive performance compared to existing models in comparative experiments when applied to Chinese corpora, English corpora, and constructed Judicial Document Corpus (JudDC), proving its adaptability, robustness, and transferability. In addition, the effectiveness of the significant components integrated attention (MHA-CA) and CRF was verified through ablation experiments, and the influence of two hyperparameters, quantity of heads and dilation rate, on the performance of model was discussed. As the key preliminary step, the proposed MDM-NER and constructed JudDR can be applied to the construction of judicial document knowledge graph, and the local deployment & data expansion tasks of vertical LLMs for judicial authorities.

1. Introduction

Named Entity Recognition (NER) is a key task in Natural Language Processing (NLP) [1,2]. It is widely used in the knowledge graphs construction [3–6], machine translation [7–9], knowledge base question answering [10,11] and information retrieval [12,13], etc. by automatically identifying and extracting entities with specific semantic meanings such as personnel, location and organization, etc. from unstructured texts. English text, with its distinct formal markings such as spaces separating words, usually has clear boundaries, making it relatively easy to label named entities [14,15]. In contrast, Chinese text is composed of continuous Chinese characters without explicit boundary identifiers, and has a more complex semantic structure, making named entity recognition more challenging [16,17]. Currently, based on technological characteristics, named entity recognition methods are mainly divided into three types: dictionary and rule based, statistical machine learning based, and deep learning based [18–20].

Specifically, dictionary and rule based named entity recognition methods typically construct the knowledge base consisting of common sense dictionary and feature words manually according to dataset characteristics, linguistic knowledge and domain expertise etc., and define rule templates that include keywords, directional words, statistical information, and indicator words, etc. Then, the entity recognition is completed by precise or fuzzy string matching method [21]. This method has the advantages of high accuracy, strong interpretability and powerful domain adaptability, but its generalization is poor, requiring labor-intensive creation and updating of dictionaries and rules.

In order to alleviate these problems, the data-driven statistical machine learning method is applied to the task. Through the statistical analysis of large-scale labeled corpora, it learns the language characteristics and distribution rules of named entities, thereby enabling automatic recognition and classification of entities in text [22]. The

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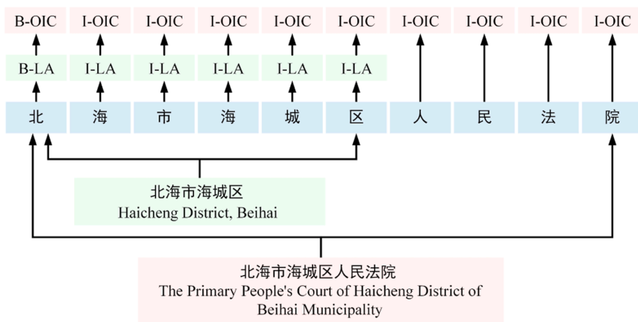

Fig. 1. Nested entities in judicial documents.

advantage lies in the fact that it can adapt to the new recognition task simply by using the corpus of the application scenario to fine tune the model. The representative machine learning methods used for named entity recognition tasks include Hidden Markov Models (HMM) [23], Conditional Random Fields (CRF) [24] and Maximum Entropy Markov Models (MEMM) [25], etc. However, the learning process of the model also requires a large quantity of labeled samples, and the performance heavily depends on feature engineering, which needs to design effective features according to domain knowledge and experience, thereby increasing the difficulty and cost of model development [26,27].

By means of its powerful feature learning capabilities, deep learning frameworks and models are widely used in the scenarios of image processing [28–30], autonomous driving [31,32], speech recognition [33,34] and sentiment analysis [35,36], etc., and have made significant breakthroughs in the task of named entity recognition [37–39]. Compared with the method based on statistical machine learning, the deep learning method does not need to manually design complex features, and can effectively capture the deep semantics and long-distance dependencies of the text, offering better robustness and transferability. At the same time, the pre-trained model will significantly reduce the training cost in low resource scenarios.

Judicial documents tend to have more complex semantic relationships, longer text descriptions and more professional terms, which brings great challenges to identify a large quantity of nested entities. Specifically, as shown in Fig. 1, “B-X” and “I-X” indicate the beginning and the intermediate elements of the X-type entity respectively, and “O” represents that it does not belong to any type. Among them, “Haicheng District, Beihai” is a location entity, and it is also nested in the organization entity “The Primary People’s Court of Haicheng District of Beihai Municipality”. The two entities have obvious differences in length and category, but exist in the same sentence. In addition, in the case shown in Fig. 2, through semantic understanding and relationship analysis of the context, it can be inferred that the entity “A fine of 30,000 yuan” belongs to the category of “Handling measures” according to the entity “Crime of infringing on citizens’ personal information” in the category of “Laws and Regulations”. At the same time, as an important data source for constructing knowledge graph of the legal domain and the vertical LLMs, there is currently a lack of corpus for judicial documents [40].

Therefore, based on the unstructured text data in judicial documents, the Judicial Document Corpus (JudDC) containing 8 entity categories, over 26 K sentences and 119 K entities is constructed, and the following contributions are made on this basis:

1. The Multiple Dependency Modeling driven Named Entity Recognition model (MDM-NER) is constructed, and its effectiveness, adaptability, performance superiority and transferability are proved by robustness test, ablation experiment and variable control;
2. To capture both local and global dependencies between characters and words, the encoder module integrates Multi-Head Attention and Cross Attention (MHA-CA) is proposed to enhance the recognition capability of nested entities;
3. The decoder module primarily composed of a joint predictor and a CRF model is designed. The joint predictor achieves multi-dimensional collaborative recognition of entities through a combination of multi-layer perceptron and Bi-affine prediction, and the introduction of CRF model is able to optimize the prediction results of entity boundaries.

2. Dataset construction

The creation of corpus is essential for the research in natural language processing, especially in the domain of legal. In this regard, the “Chinese Judgment Online” (https://wenshu.court.gov.cn) was served as a key resource, offering a large collection of case information. This research concentrates on the cases in the aspect of data security and perform detailed extraction and analysis of criminality, including the infringement of citizens’ personal information and the unauthorized access to data.

The automated data collection system was developed based on web crawling technology and extract the unstructured text data of 2,300 legal documents. The operations of standardizing date formats and removing special characters etc. are performed to improve the consistency and usability of data. On this basis, the categories of entities in the texts were manually annotated, resulting in a judicial document corpus

Table 1
Comparison of different corpora.

Datasets	GENIA [41]	CoNLL03 [42]	MSRA [43]	Weibo [44]	Resume [45]	JudDC
Quantity of entity types	5	4	3	4	**8**	**8**
Quantity of sentences	18,545	22,137	**50,729**	1,890	4,761	26,155
Quantity of entities	56,870	35,089	80,882	1,981	16,565	**119,578**
Average sentence length	23.56	13.62	46.18	54.52	32.15	**61.29**

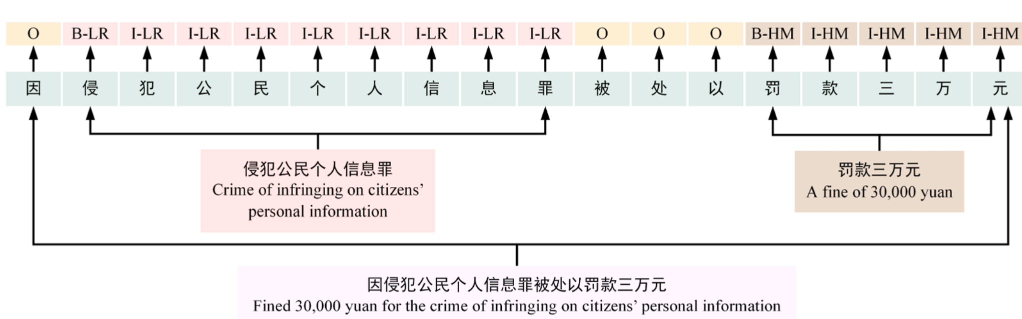

Fig. 2. Entity type inference based on context.

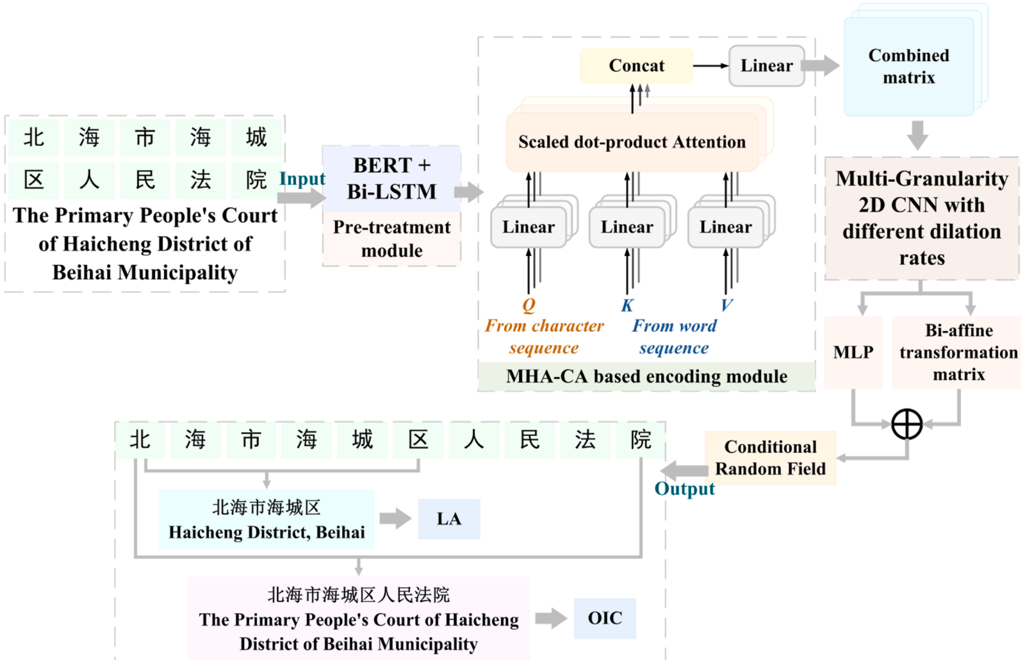

Fig. 3. The architecture of MDM-NER.

Table 2
The experiment results of different models on JudDC.

Model	JudDC		
	P	*R*	*F*1
TENER [48]	0.6987	0.7872	0.7403
W²NER [49]	0.7032	0.8054	0.7508
DIFFUSIONNER [50]	0.6872	0.7615	0.7224
MDM-NER	**0.7578**	**0.8628**	**0.8069**

named JudDC containing 26,155 sentences and 119,578 entities. Among them, the 8 entity categories are laws and regulations (LR), handling measures (HM), personnel involved in the case (PC), organizations and institutions involved in the case (OIC), location and address (LA), date and time (DT), judicial authorities and staff (JAS), and the subject matter of the case (SMC). JudDC will not only be capable of supporting the training and verification of NER networks, but also providing the data foundation and factual basis for the domain knowledge graph construction of judicial documents and the local deployment & fine-tuning of vertical LLMs for judicial authorities.

As shown in Table 1, the constructed well-balanced JudDC has relatively more entity type and scale, and more complex content compared with existing the public corpora. Within these, GENIA is a

Table 3
The comparative experiment results on Chinese datasets.

Model	MSRA			Resume			Weibo		
	P	*R*	*F*1	*P*	*R*	*F*1	*P*	*R*	*F*1
TENER [48]	–	–	0.9274	–	–	0.9500	–	–	0.5817
LGN [51]	0.9450	0.9293	0.9371	0.9537	0.9485	0.9511	0.5714	0.6677	0.6158
SoftLexicon [52]	0.9575	0.9510	0.9542	0.9609	0.9613	0.9611	0.7094	0.6702	0.6892
Locate [53]	0.9221	0.9072	0.9146	–	–	–	0.7011	0.6812	0.6910
W²NER [49]	0.9488	0.9506	0.9497	**0.9655**	0.9627	**0.9641**	0.7094	**0.7387**	0.7238
DIFFUSIONNER [50]	0.9571	0.9412	0.9491	0.9521	0.9509	0.9515	0.6520	0.6787	0.6651
MDM-NER	**0.9584**	**0.9573**	**0.9578**	0.9604	**0.9644**	0.9624	**0.7221**	0.7343	**0.7281**

Table 4
The comparative experiment results on English datasets.

Model	GENIA			CoNLL03		
	P	*R*	*F*1	*P*	*R*	*F*1
Biaffine [54]	0.8180	0.7930	0.8053	0.9291	0.9213	0.9252
BERT-MRC [55]	0.8114	0.7682	0.7892	0.9233	**0.9376**	0.9304
BARTNER [56]	0.7887	0.7960	0.7923	0.9257	0.9356	0.9306
BioNLP [57]	0.7792	0.8074	0.7930	–	–	–
Triaffine [58]	0.8042	**0.8206**	0.8123	–	–	–
W²NER [49]	**0.8310**	0.7975	0.8139	0.9271	0.9344	0.9307
DIFFUSIONNER [50]	0.8210	0.8097	0.8153	**0.9299**	0.9256	0.9277
MDM-NER	0.8230	0.8083	**0.8156**	0.9297	0.9342	**0.9319**

Table 5
Ablation experiment for each component in MDM-NER.

Model	JudDC $F1$	GENIA
w/o MHA	0.8032	0.8135
w/o CA	0.7804	0.8137
w/o MHA-CA	0.7582	0.8122
w/o CRF	0.7965	0.8145
MDM-NER	**0.8069**	**0.8156**

Table 6
The influence of the quantity of attention head on the performance of the model.

Quantity of attention heads	JudDC $F1$	Weibo	Resume	GENIA	CoNLL03	MSRA
$N = 4$	0.8014	0.6987	0.9553	0.8134	0.9245	0.9481
$N = 8$	**0.8069**	**0.7281**	**0.9624**	**0.8156**	**0.9319**	**0.9578**
$N = 16$	0.7848	0.7172	0.9610	0.8069	0.9208	0.9566
$N = 32$	0.7551	0.6980	0.9582	0.8119	0.9229	0.9573
$N = 64$	0.7789	0.6985	0.9593	0.8100	0.9265	0.9544

semantically annotated biological literature corpus for bioinformatics [46,47]; The corpus of CoNLL03 is taken from Reuters news stories; MSRA is a simplified character corpora provided by Microsoft Research Asia; Weibo is a corpus of messages annotated for both name and nominal mentions; Resume is collected from Sina Finance, which contains resumes of senior executives from listed companies in the Chinese stock market.

3. Model architecture

In order to achieve accurate recognition of multiple categories and large quantity of nested entities in highly specialized and semantically complex judicial documents, the Multiple Dependency Modeling driven Named Entity Recognition model (MDM-NER) is proposed as shown in Fig. 3. The introduction of the pre-trained language model BERT and Bi-LSTM aims to learn the expression patterns and terminology characteristics of judicial documents, and the captured long-distance semantic dependencies in the text are taken as the input for the encoder module. The multi-head attention and cross attention are integrated to capture the global dependencies of documents. After the dilated convolution operation, the multi-dimensional collaborative entity recognition is achieved through the joint predictor composed of multi-layer perceptron and Bi-affine prediction. Ultimately, CRF is employed to optimize the label sequence.

3.1. Encoder module

In view of the characteristics of nested entities spanning a large range in the judgment documents, the association relationships between characters and words are captured by the coding module which integrates Multi-Head Attention and Cross Attention (MHA-CA), thereby improving the recognition ability of the proposed model for nested entities. The application of integrated attention to independent heads can be calculated with Eq. (1):

$$MHA - CA\left(Q_{c_n}, K_{w_n}, V_{w_n}\right) = Softmax\left(\frac{Q_{c_n} {K_{w_n}}^T}{\sqrt{d}}\right) V_{w_n} \tag{1}$$

where Q_{c_n} is derived from the character sequence, and K_{w_n} and V_{w_n} are

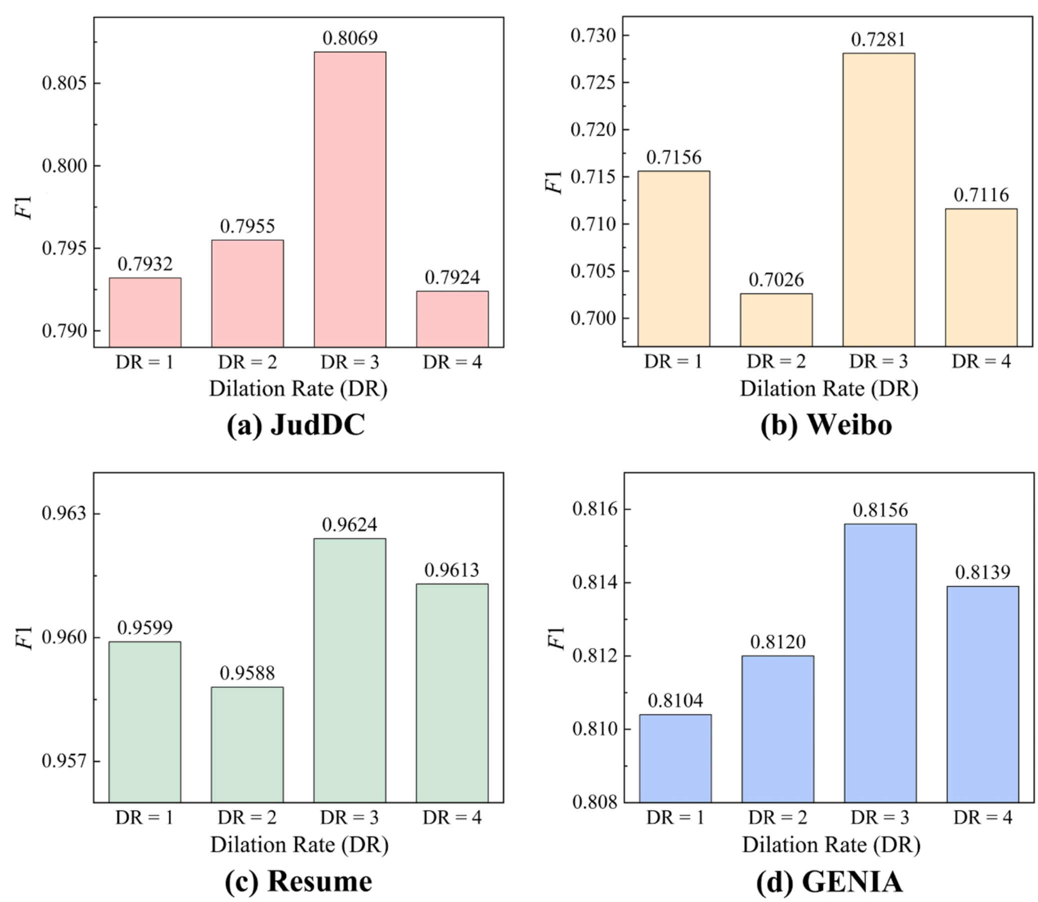

Fig. 4. The performances of models with different dilation rates.

both derived from the word sequence, n indicates that the current calculation is directed toward the n-th head. When extended to multiple attention heads, each head is able to independently learn local dependencies between characters and words, and capture the global dependencies by applying learnable weighting mechanisms to different heads, thus enhancing the representation ability of the model for nested entities in complex semantic structures. The result *MH_Output* after the corresponding operations to all heads is shown in Eq. (2):

$$MH\_Output = Concat\left(LW\left(head^{(1)}, head^{(2)}, \ldots, head^{(j)}, \ldots, head^{(i)}\right)\right) \tag{2}$$

where $head^{(j)}$ represents the output of the j-th head, i denotes the quantity of heads, LW and $Concat$ denote the weighting and concatenation operations, respectively. After that, the normalization layer is introduced to improve the model's training efficiency and generalization ability of the, and the Multi-Granularity 2D CNN with different dilation rates is utilized to further capture the multi-scale feature representation.

3.2. Decoder module

The decoder module is mainly composed of joint predictor and CRF model. Among them, the joint predictor realizes multi-dimensional collaborative recognition of nested entities through the combination of Multi-Layer Perceptron (MLP) and Bi-affine prediction. Specifically, by continuously updating the weights and biases during training, the MLP is able to complete the mapping of feature representation to named entity labels. The Bi-affine prediction applies a nonlinear activation function after the calculation of the Bi-affine transformation matrix to generate the prediction result, and finally obtains the output results of the joint predictor through the *Softmax* function shown in Eq. (3):

$$JP\_r = Softmax(\mathrm{MLP\_m} + BA\_p) \tag{3}$$

where JP_r is the result of the joint prediction, MLP_m and BA_p are the outputs of Multi-Layer Perceptron and Bi-affine predictor, respectively. The CRF model is capable of reducing incorrect entity boundary predictions based on the transition probabilities between labels, thereby generating more reasonable label sequences and improving the accuracy and robustness of the model. Therefore, after the joint prediction operation, the CRF model shown in Eq. (4) is used to further optimize the prediction results:

$$P(RELS|I\_T) = \frac{exp(\sum_{Sentence} W_a \times TRPR + W_b \times JP\_r)}{\sum_{APELS} exp(\sum_{Sentence} W_a \times TRPR + W_b \times JP\_r)} \tag{4}$$

among them, W_a and W_b are trainable weight parameters, I_T and $RELS$ are the input text and the real entity label sequence, respectively. $APELS$ represents all possible entity label sequences, and $TRPR$ is the transition probability matrix.

4. Experiment results

4.1. Experiment setting and metrics

The effectiveness of the proposed MDM-NER is demonstrated by comparing its performances with existing models applied to Chinese and English benchmark corpora. Specifically, the Chinese corpora include MSRA [43], Weibo [44], Resume [45], and JudDC, while the English corpora include GENIA [41] and CoNLL03 [42]. Each of them is divided into training, validation, and test sets in an 8:1:1 ratio, and the cross-entropy loss function is selected for model parameter update. The utilized deep learning framework was TensorFlow 2.8, the version of Python was 3.9, and the parallel computing architecture was CUDA 11.2. The dimension of word embedding is 768, the learning rate and epoch are 0.001 and 100, respectively. Moreover, the proposed model can be trained on corpora from various domains to adapt to recognition tasks in more application scenarios.

The evaluation metrics shown in Eqs. (5-7) are used to measure the performance of the model:

$$P = \frac{TP}{TP + FP} \tag{5}$$

$$R = \frac{TP}{TP + FN} \tag{6}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{7}$$

where TP denotes the true positive sample, FP and FN represent the false positive sample and the false negative sample, respectively.

4.2. Overall performance and robustness evaluation

Table 2 shows the advantages of MDM-NER when applied to named entity recognition tasks on JudDC. The proposed model is superior to other models in P, R and $F1$, which proves its effectiveness in dealing with complex texts containing a large quantity of nested entities in judicial documents.

Furthermore, we extend MDM-NER to three Chinese corpora and compare it with seven existing methods, as shown in Table 3. The results indicate that MDM-NER outperforms the existing models in terms of P, R, and $F1$ by at least 0.09%, 0.63% and 0.0036 respectively when applied to MSRA, demonstrating its applicability in the texts of news domain. Simultaneously, it achieves a $F1$ second only to W^{2}NER on Resume, and demonstrates certain advantages over other methods in terms of P and $F1$ when tests on the Weibo corpus, which is oriented toward the social media, thereby showcasing its strong domain transferability.

As shown in Table 4, the proposed method is compared with 7 existing researches in English corpus. It achieved higher $F1$ scores on both GENIA and CoNLL03, indicating that MDM-NER has better comprehensive performance. Meanwhile, the adaptability and robustness of the model are also proved by its ability to adapt to both Chinese and English corpora.

4.3. Ablation experiments and variable control

For the purpose of proving the effectiveness of each component in MDM-NER, ablation experiments shown in Table 5 are carried out. The results show that all components have a positive effect on the model's performance in the recognition of nested entities. Specifically, on the constructed JudDC, CA has the greatest impact on the model as a single factor, while on the GENIA, MHA and CA have similar influence, and CRF can relatively slightly enhance the model's recognition capabilities. Additionally, when the integrated attention is absent, the model's performance is the poorest, indicating a synergistic enhancement effect among the components.

In the encoder module, the quantity of heads is an important hyperparameter. Less heads directly result in insufficient learning ability for complex dependencies, while more heads mean greater computational overhead. Therefore, the impact of different head quantities on the performance of model is analyzed based on Chinese and English corpora, as shown in Table 6.

The experiment results indicate that when the quantity of heads is 8, the model demonstrates relatively superior performance across all datasets. When the quantity of head is small, the model's performance may be constrained by the feature space, leading to the lack of expression capability. Conversely, more heads may cause over fitting, resulting in the degradation of the model.

The dilation rate directly determines the receptive field of the convolution kernel, which in turn affects the granularity of features after 2D convolution operations. Therefore, the relatively optimal choice is explored by setting different dilation rates as shown in Fig. 4. The experiment results show that all datasets exhibit the best performance when the value is 3, which proves that it has the relatively most appropriate sampling interval.

5. Conclusion and discussion

To achieve the recognition of nested entities in judicial documents, this paper constructed a judicial document corpus with the name of JudDC, containing over 26 K sentences and 119 K entities. On this basis, the multiple dependency modeling driven named entity recognition model is proposed. When applied to Chinese and English corpora, the better comprehensive performance of MDM-NER compared with the existing models proves its adaptability, robustness and transferability. At the same time, the ablation experiments proved the effectiveness of the significant components MHA-CA and CRF, and discussed the influence of the quantity of heads and dilation rate on the performance of the model. The proposed recognition method is the key preliminary step in constructing knowledge graph of judicial documents, and it can also provide an important theoretical foundation and technical reference for the local deployment and data expansion of vertical LLMs in the domain of legal.

CRediT authorship contribution statement

Haoze Yu: Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qunhe Ji:** Visualization, Methodology. **Yan Li:** Software, Data curation. **Huanpu Yin:** Methodology, Investigation. **Haisheng Li:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Funding acquisition. **Xiaohui Li:** Methodology, Funding acquisition, Conceptualization. **Junping Du:** Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.inffus.2025.103807.

Data availability

No data was used for the research described in the article.

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